Orbit-resolved spectra and dynamical accessibility in multi-register covering Hamiltonians

Fabrício de Souza Luiz

Instituto de Física Gleb Wataghin, Universidade Estadual de Campinas, 13083-859 Campinas, SP, Brazil

August 11, 2026

Which part of a Hamiltonian spectrum is physically relevant when the initial state and the interpolation preserve several symmetries? We answer this question for a multi-register encoding of fixed-cardinality Set Cover. The joint action combines register permutations with the faithful base action of the incidence automorphism group. Its orbits are unlabelled candidate covers, and the exact orbit quotient is a hopping Hamiltonian on those candidates. We prove that the classical penalties are orbit invariants and distinguish the resulting fixed space from the generally smaller cyclic space generated by the protocol. A sector-resolved Schur-complement bound then certifies cover-measurement probability with the kinetic floor required by sector projection. For the retained-walk linear path, a general hopping-cancellation point makes the spectrum exactly orbit-resolved; multiplicities outside the fixed sector are dynamically dark, whereas transverse dark crossings obey a separate stability theorem. On even cycles the fixed-sector excitation gap at the cancellation point is $\lambda(n-1)/(2n-1)$, where λ is the uncovered-task penalty, despite the global multiplicity closure. Separately, on the same instance family, we construct a different Johnson–Metropolis parent path from a Dicke state to a Gibbs-amplitude state; this second construction controls an entire accessible path rather than a single cancellation point. Its cover-failure probability is bounded by a constant times n^{-5} and its cyclic gap is bounded below by $1024\,n^{-13}$ uniformly along the path. Quantitative derivative bounds yield a conditional polynomial adiabatic runtime in the abstract Hamiltonian-access model. We make no quantum-speedup claim: the result is a protocol-dependent separation of global, symmetry-fixed, and dynamically accessible spectra.

Fabrício de Souza Luiz: fsluiz@unicamp.br

1 Introduction

Symmetry block diagonalizes a Hamiltonian, but block diagonalization alone does not determine which eigenstates a protocol can occupy. If an initial state is fixed by every preserved symmetry, evolution is confined to the joint fixed space. Even that statement can be loose: the algebra generated by the endpoint Hamiltonians may act cyclically only on a proper subspace of the joint fixed space. A small global gap can then describe a crossing that the protocol neither couples to nor detects.

Set Cover, whose minimum-cardinality version is NP-hard [1], admits the usual one-binary-variable-per-base Ising formulation, and it also appears among the constrained examples motivating quantum alternating-operator ansatzes [2, 3]. Separately, qubit-efficient variational encodings show that logarithmic or otherwise compressed register counts are not by themselves a novelty claim [4–6]. Our multi-register address encoding belongs to this broad line; no result below depends on claiming a new qubit count.

Classical objective symmetries can confine quantum approximate optimization algorithm (QAOA) dynamics and impose equal measurement probabilities on symmetry-related strings [7]. Quantum walks on automorphism quotients likewise identify a smaller invariant evolution for suitable initial states [8], while Krylov methods use repeated generator action to identify a minimal dynamical subspace [9]. These ideas motivate, but do not identify with one another, the two spaces used below: the group-fixed orbit space and the generally smaller cyclic space of the full protocol algebra.

Finally, symmetric discriminants of reversible Markov chains and annealing paths with Gibbs-amplitude ground states are established tools in quantum walk and quantum simulated-annealing constructions [10, 11]. Quantitative adiabatic computation originates with the interpolation framework of Farhi et al. [12]; modern reviews emphasize that runtime claims require a specified path and access model [13]. Quantitative statements require an isolated spectral band together with derivative and gap control [14]. We use the discriminant only to construct a rigorously gapped witness path and make no speedup claim.

Minimum Set Cover (MSC) provides a concrete setting in which these three spaces differ. With n_B candidate bases, a k-register encoding represents an ordered candidate by $k\lceil\log_2 n_B\rceil$ address qubits. Permuting the registers gives an action of the symmetric group S_k, while colour-preserving automorphisms of the incidence relation permute the base labels. Register symmetry alone and a final-Hamiltonian sector gap are insufficient for an adiabatic or otherwise path-dependent statement: one must use the faithful joint action, the protocol's cyclic space, and the minimum isolation gap along an explicit path.

This work makes that distinction its central question. Its three main contributions are:

1. an orbit description of the k-register encoding under $S_k \times G_B$, where G_B is the faithfully represented group of base automorphisms; this gives invariant covering penalties, exact Pólya–Redfield counts, and a constructive distinction between the fixed space $\mathcal{H}_{\mathrm{sym}}$ and the protocol-cyclic space $\mathcal{K}_u$;
2. a sector-localization proposition with its kinetic floor, together with a general orbit-resolved hopping-cancellation theorem and a separate stability theorem for transverse inter-sector crossings; and
3. an even-cycle illustration of the cancellation theorem and, separately, a Johnson–Metropolis parent protocol whose accessible gap is uniformly polynomial, yielding a conditional polynomial-time adiabatic corollary.

The last result is purposely about accessibility rather than computational advantage. The controlled cycle family is classically easy, and the proven exponent is conservative. Its role is to separate the gap question cleanly into symmetry, geometry, and kinetics, without inferring asymptotic behavior from finite-size data.

Neither orbit reduction, Pólya enumeration, compact addressing, nor a Markov discriminant is claimed as new in isolation. The contribution is the protocol-dependent link between orbit potentials, group-fixed spectra, cyclic closure, sector localization, and dark multiplicity, together with a separately controlled parent path.

We use the following notation throughout. For an operator X, $\operatorname{Ran} X$ and $\operatorname{spec} X$ denote its range and spectrum. The symbol $\|\cdot\|$ denotes the Hilbert-space norm on vectors and the induced operator norm on operators. The gap of a finite-dimensional Hermitian operator is the distance from its lowest eigenspace to the next distinct eigenvalue; $\lambda_{\min}(X|_{\mathcal{L}})$ is the lowest eigenvalue of the compression of X to $\mathcal{L}$. For positive g, the notation $f = O(g)$ means $|f| \le Cg$ for a constant $C > 0$ independent of the asymptotic variable in the stated limit.

2 Encoding, orbit structure, and cyclic accessibility

Let $\mathcal{T} = \{1, \dots, n_T\}$ be a task universe and let $\mathcal{B}_b \subseteq \mathcal{T}$, $1 \le b \le n_B$, be the candidate bases. Equivalently, $A \in \{0,1\}^{n_B \times n_T}$ is their incidence matrix, with $A_{bt} = 1$ exactly when $t \in \mathcal{B}_b$. Minimum Set Cover asks for

$$k^\star = \min\{|S| : S \subseteq [n_B], \quad \textstyle\bigcup_{b\in S} \mathcal{B}_b = \mathcal{T}\}. \tag{1}$$

We fix $1 \le k \le n_B$ address registers and first work in the physical address space

$$\mathcal{H}_{\mathrm{addr}} = (\mathbb{C}^{n_B})^{\otimes k}. \tag{2}$$

The embedding into $2^{\lceil\log_2 n_B\rceil}$ levels and its dynamically disconnected padding block are implementation details.

Write $|\boldsymbol{b}\rangle = |b_1\rangle \otimes \cdots \otimes |b_k\rangle$ for a computational-basis tuple. Its numbers of uncovered tasks and repeated base pairs are

$$U(\boldsymbol{b}) = \sum_{t=1}^{n_T} \prod_{r=1}^{k} (1 - A_{b_r,t}), \tag{3}$$

$$D(\boldsymbol{b}) = \sum_{1\le r<s\le k} \mathbf{1}[b_r = b_s], \tag{4}$$

respectively, where $\mathbf{1}[\cdot]$ is the indicator of the enclosed condition. Thus $U(\boldsymbol{b}) = D(\boldsymbol{b}) = 0$ exactly when a measurement returns an exact-k cover with distinct bases. When $k = k^\star$, every such outcome solves MSC; for other k, the Hamiltonian encodes the fixed-cardinality covering problem rather than minimum cardinality itself.

Let $|e\rangle = \sum_{b=1}^{n_B} |b\rangle$ and let I_B be the identity on one address register. The normalized complete-graph hopping matrix and its k-register lift are

$$W = \frac{|e\rangle\langle e| - I_B}{n_B - 1}, \qquad H_{\mathrm{walk}} = \sum_{r=1}^{k} W^{(r)}, \tag{5}$$

where $W^{(r)}$ acts as W on register r and as the identity on all others. Define the diagonal penalty operators

$$H_{\mathrm{cov}} = \sum_{\boldsymbol{b}} U(\boldsymbol{b}) |\boldsymbol{b}\rangle\langle\boldsymbol{b}|, \qquad H_{\mathrm{excl}} = \sum_{\boldsymbol{b}} D(\boldsymbol{b}) |\boldsymbol{b}\rangle\langle\boldsymbol{b}|. \tag{6}$$

Both sums run over all n_B^k computational-basis tuples. For positive penalty energies λ and μ, the problem Hamiltonian is

$$H_{\mathrm{prob}} = H_{\mathrm{walk}} + \lambda H_{\mathrm{cov}} + \mu H_{\mathrm{excl}}, \tag{7}$$

so $\lambda U(\boldsymbol{b}) + \mu D(\boldsymbol{b})$ is the total diagonal penalty of $|\boldsymbol{b}\rangle$. An explicit initial Hamiltonian is

$$H_{\mathrm{init}} = \sum_{r=1}^{k} \left(I_B - |u_B\rangle\langle u_B|\right)_r, \qquad |u_B\rangle = n_B^{-1/2} \sum_{b=1}^{n_B} |b\rangle, \tag{8}$$

with unique ground state $|u\rangle = |u_B\rangle^{\otimes k}$ and initial gap one. The retained-walk linear interpolation is parameterized by $s \in [0, 1]$ as

$$H(s) = (1 - s)H_{\text{init}} + sH_{\text{prob}}. \tag{9}$$

2.1 The faithfully represented group

The colour-preserving incidence automorphism group is

$$G_A = \{(\pi_B, \pi_T) \in S_{n_B} \times S_{n_T} : A_{\pi_B(b),\pi_T(t)} = A_{bt} \text{ for all } b, t\}, \tag{10}$$

where S_d denotes the symmetric group on d labels. Only its base component acts on Eq. (2). We therefore use the effective group

$$G_B = \mathrm{pr}_B(G_A) \simeq G_A/\ker\rho, \qquad \rho(\pi_B, \pi_T) = \pi_B, \tag{11}$$

where pr_B projects an automorphism onto its base permutation and $\ker\rho$ contains the automorphisms acting trivially on all bases. Thus G_B is the faithfully represented base group. The register and base actions commute, so the represented symmetry is

$$G = S_k \times G_B. \tag{12}$$

Every term in Eqs. (7) and (8) commutes with G. Consequently, any Hamiltonian H commuting with G has the block structure

$$\mathcal{H}_{\text{addr}} \simeq \bigoplus_{\alpha,\beta} V_\alpha^{S_k} \otimes V_\beta^{G_B} \otimes \mathcal{M}_{\alpha\beta}, \tag{13}$$

$$H \simeq \bigoplus_{\alpha,\beta} I_{V_\alpha} \otimes I_{V_\beta} \otimes h_{\alpha\beta}. \tag{14}$$

Here α and β label irreducible representations of S_k and G_B; the spaces $V_\alpha^{S_k}$ and $V_\beta^{G_B}$ carry those representations; $\mathcal{M}_{\alpha\beta}$ is their multiplicity space; and $h_{\alpha\beta}$ is the reduced action on that space. The symbols I_{V_α} and I_{V_β} denote the corresponding identities. This is the source of Schur multiplicities [15]; an equality of eigenvalues in different blocks is a separate phenomenon. The full representation and the role of the kernel of G_A are detailed in Appendix B.

2.2 Allowed space versus accessible space

Let U_σ permute the k registers according to $\sigma \in S_k$, and let V_g apply the base permutation $g \in G_B$ in every register. The joint fixed projector and its range are

$$\Pi_{\text{sym}} = \left(\frac{1}{k!}\sum_{\sigma\in S_k} U_\sigma\right)\left(\frac{1}{|G_B|}\sum_{g\in G_B} V_g\right), \tag{15}$$

$$\mathcal{H}_{\text{sym}} = \operatorname{Ran}\Pi_{\text{sym}}. \tag{16}$$

This is the symmetry-allowed invariant space, not automatically the space filled by the protocol. Let $\mathfrak{A} = \mathrm{alg}^*(H_{\text{init}}, H_{\text{prob}}, I)$ be the smallest unital algebra containing the endpoints and closed under adjoints; here I is the identity on $\mathcal{H}_{\text{addr}}$. The dynamically accessible cyclic space is

$$\begin{aligned}\mathcal{K}_u &= \mathfrak{A}|u\rangle \\ &= \operatorname{span}\{X_\ell \cdots X_1|u\rangle : X_i \in \{H_{\text{init}}, H_{\text{prob}}\},\ \ell \geq 0\}.\end{aligned} \tag{17}$$

It is the smallest common reducing subspace that contains the initial state, and in general

$$\mathcal{K}_u \subseteq \mathcal{H}_{\text{sym}} \subseteq \mathcal{H}_{\text{addr}}. \tag{18}$$

The first inclusion can be strict; padding makes it strict automatically if one defines the symmetry on the full qubit space.

Figure 1 schematically summarizes the two distinctions used throughout the paper. Group invariance identifies an allowed space, while the initial state and generator algebra select the smaller cyclic space. An eigenvalue coincidence in another irrep may then close the global multiplicity gap without closing the isolation gap seen by the protocol.

2.3 Orbit invariants and exact quotient

For $\boldsymbol{b} = (b_1, \ldots, b_k)$, the joint action is

$$(\sigma, g)\boldsymbol{b} = \big(g(b_{\sigma^{-1}(1)}), \ldots, g(b_{\sigma^{-1}(k)})\big), \quad (\sigma, g) \in S_k \times G_B. \tag{19}$$

The following elementary observation connects this representation directly to the covering problem.

Lemma 1 (Orbit invariance of the covering penalties)**.** *The functions U and D are constant on every orbit of the action in Eq. (19). Consequently,*

$$\Phi(\boldsymbol{b}) = \lambda U(\boldsymbol{b}) + \mu D(\boldsymbol{b}) \tag{20}$$

is an orbit function, the feasible set $\{U = D = 0\}$ is a union of orbits, and its projector P commutes with G.

A register-permutation orbit is a size-k multiset of bases. A joint G-orbit is therefore a G_B-equivalence class of such multisets: an unlabelled candidate k-cover, which may still be infeasible or contain repetitions. The Cauchy–Frobenius lemma and its Pólya–Redfield cycle-index formulation [16–18] give their number as

Proposition 2 (Orbit classification and count)**.** *Let $c_\ell(g)$ be the number of length-ℓ cycles of $g \in G_B$ in its base action. Then*

$$\dim \mathcal{H}_{\text{sym}} = \frac{1}{|G_B|}\sum_{g\in G_B} [z^k] \prod_{\ell\geq 1} (1 - z^\ell)^{-c_\ell(g)}. \tag{21}$$

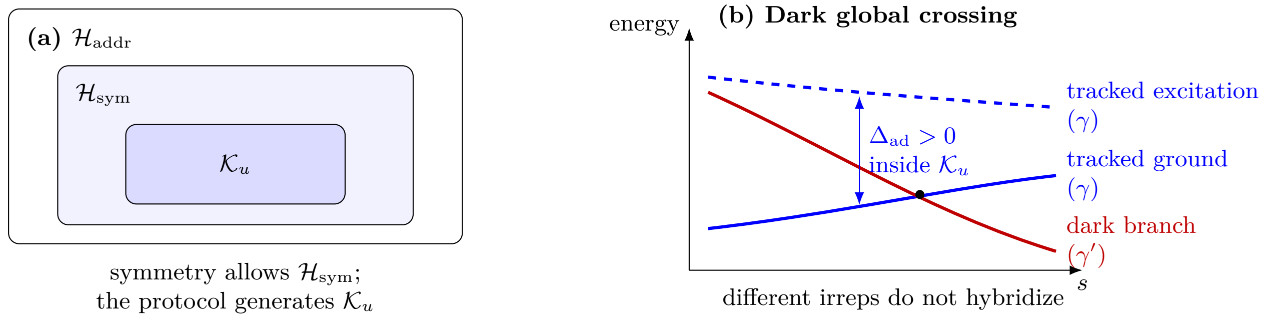

Figure 1: Protocol-dependent spectral relevance. (a) The joint-fixed space is only symmetry allowed; the cyclic space also depends on the initial state and generated algebra. (b) The labels γ and γ' denote inequivalent symmetry sectors. A branch in the latter can cross the followed branch and close a global multiplicity gap while the tracked band remains isolated within the cyclic space. Solid blue is the tracked ground branch, dashed blue its in-sector excitation, and red an inequivalent dark branch. The diagram is not to scale.

Here $[z^k]$ extracts the coefficient of z^k. Equivalently, the right-hand side is the cycle index of the base action evaluated on the generating functions for multisets.

Equation (21) counts basis states in the group-fixed space, not all vectors in the full quantum Hilbert space and not all states that a particular protocol can prepare. The latter question requires the cyclic space below.

Orbit quotients of symmetry-confined quantum walks are established constructions [8]; we use the joint action here to retain the exact Hamiltonian matrix elements. For an orbit $\mathcal{O}$, define

$$|\mathcal{O}\rangle = |\mathcal{O}|^{-1/2} \sum_{x\in\mathcal{O}} |x\rangle. \tag{22}$$

These vectors are an orthonormal basis of $\mathcal{H}_{\rm sym}$. If H commutes with G, $x \in \mathcal{O}$, and $H_{yx} := \langle y|H|x\rangle$, then

$$\langle \mathcal{O}'|H|\mathcal{O}\rangle = \sqrt{\frac{|\mathcal{O}|}{|\mathcal{O}'|}} \sum_{y\in\mathcal{O}'} H_{yx}. \tag{23}$$

The sum is independent of the representative x. Lemma 1 makes the penalty term diagonal in this basis:

$$\langle \mathcal{O}'|V|\mathcal{O}\rangle = \delta_{\mathcal{O}\mathcal{O}'}\Phi(\mathcal{O}). \tag{24}$$

Thus the quotient is a single-particle hopping problem. Its sites are unlabelled candidate covers, Φ is the on-site potential, and the off-diagonal entries of Eq. (23) are hopping amplitudes with orbit-multiplicity factors. The graph is connected because one-register changes connect any two size-k multisets. Reachability of a solution site is therefore not the obstruction: the protocol must redistribute the state's amplitude into the fixed solution subspace

$$\operatorname{Ran} P \cap \mathcal{H}_{\rm sym} = \operatorname{span}\{|\mathcal{O}\rangle : \Phi(\mathcal{O}) = 0\}. \tag{25}$$

An individual computational solution need not be group fixed, but its uniform orbit state always lies in Eq. (25). Nontrivial combinations within the same solution orbit belong to other irreps and can be inaccessible from $|u\rangle$. Away from a hopping-cancellation point, orbit potentials are not spectral gaps because eigenvectors are generally delocalized across several sites. Moreover, nontrivial irreps have no site in this quotient; this is precisely why inter-sector partners of dark crossings are absent from the dynamics shown by the orbit graph.

2.4 Cyclic accessibility in orbit coordinates

Let

$$M = \sum_{r=1}^{k} |e\rangle\langle e|_{(r)}, \qquad V = \lambda H_{\rm cov} + \mu H_{\rm excl}. \tag{26}$$

The endpoint algebra has only these two nontrivial generators.

Lemma 3 (Two-generator reduction)**.**

$$\operatorname{alg}^*(H_{\rm init}, H_{\rm prob}, I) = \operatorname{alg}^*(M, V, I). \tag{27}$$

Let $\widehat{M}$ and $\widehat{V}$ be the orbit quotients of M and V. Write $\varphi_1 < \cdots < \varphi_p$ for the distinct values of Φ on the orbits and let R_j project onto the corresponding potential shell. The initial vector has strictly positive orbit coordinates

$$w_{\mathcal{O}} = \langle \mathcal{O}|u\rangle = \sqrt{|\mathcal{O}|/n_B^k} > 0. \tag{28}$$

Thus every orbit has nonzero amplitude initially, but this does not imply that the individual basis vector $|\mathcal{O}\rangle$ can be prepared by the protocol. A strict cyclic inclusion excludes linear combinations—patterns of relative amplitudes and phases across orbit sites—rather than deleting particular sites from the graph.

Theorem 4 (Shell-resolved cyclic closure)**.** *In orbit coordinates, $\mathcal{K}_u$ is the smallest subspace containing w and invariant under $\widehat{M}$ and $\widehat{V}$. It is obtained by the* *stabilising iteration*

$$\mathcal{K}^{(0)} = \operatorname{span}\{R_j w : 1 \le j \le p\}, \tag{29}$$

$$\mathcal{K}^{(r+1)} = \mathcal{K}^{(r)} + \sum_{j=1}^{p} R_j \widehat{M} \mathcal{K}^{(r)}. \tag{30}$$

Moreover, the accessible and inaccessible directions decompose shell by shell:

$$\mathcal{K}_u = \bigoplus_{j=1}^{p} R_j \mathcal{K}_u, \tag{31}$$

$$\mathcal{K}_u^{\perp} \cap \mathcal{H}_{\text{sym}} = \bigoplus_{j=1}^{p} \left(\operatorname{Ran} R_j \ominus R_j \mathcal{K}_u\right). \tag{32}$$

Consequently,

$$\dim \mathcal{K}_u \ge p, \tag{33}$$

and $\mathcal{K}_u = \mathcal{H}_{\text{sym}}$ whenever Φ is injective on orbits. Conversely, a strict inclusion $\mathcal{K}_u \subsetneq \mathcal{H}_{\text{sym}}$ requires two distinct orbits to share a potential value.

For the linear path in Eq. (9), let

$$c(s) = \frac{s}{n_B - 1} - \frac{1-s}{n_B}. \tag{34}$$

Whenever $c(s) \le 0$, the symmetric and cyclic ground energies agree:

$$E_0^{\mathcal{K}}(s) = E_0^{\text{sym}}(s). \tag{35}$$

The last statement follows physically from the sign-free, connected orbit hopping problem. Before cancellation, Perron–Frobenius [19] gives a unique strictly positive symmetric ground vector, which has nonzero overlap with w and hence belongs to the cyclic space. At cancellation, a nonzero projection of w remains in the lowest potential shell. Strict dimension reduction therefore need not incur a ground-energy cost; the numerical audit in Section 6 verifies the theorem on the path grid through s_c. At the problem endpoint $s = 1$, outside the theorem's sign regime, four of the five strict inclusions still have zero cyclic cost empirically. Proofs of the orbit statements and Theorem 4 are given in Appendix B.

3 Sector localization and relevant gaps

Let P project onto the span of computational-basis tuples that use distinct bases and cover every task, and put $Q = I - P$. By Lemma 1, both projectors commute with the joint group. Write $V = \lambda H_{\text{cov}} + \mu H_{\text{excl}}$, $m = \min\{\lambda, \mu\}$ for the minimum single-constraint penalty, and $\kappa = k/(n_B - 1)$ for the magnitude of the lowest possible k-register walk energy. Here k is the register count fixed in Section 2. Then $VP = 0$ and the complete-graph normalization gives

$$QH_{\text{prob}}Q \ge (m - \kappa)Q. \tag{36}$$

The global localization statement cannot be copied verbatim into an arbitrary sector. A projected cover is a superposition of computational covers and can have nonzero kinetic expectation. The missing term is the sector's kinetic floor.

Proposition 5 (Sector localization)**.** *Let $\gamma = (\alpha, \beta)$ label an isotypic component in the joint decomposition of Eq. (14), or more generally let Π_γ be any reducing sector projector commuting with H_{prob}, H_{walk}, V, and P. Suppose $P_\gamma = P\Pi_\gamma \ne 0$. Define the invalid-sector projector Q_γ, the feasible-to-invalid coupling B_γ, and the sector kinetic floor η_γ by*

$$Q_\gamma = Q\Pi_\gamma, \qquad B_\gamma = Q_\gamma H_{\text{walk}} P_\gamma, \tag{37}$$

$$\eta_\gamma = \lambda_{\min}\left(P_\gamma H_{\text{walk}} P_\gamma|_{\operatorname{Ran} P_\gamma}\right). \tag{38}$$

If $Q_\gamma = 0$, every state in the sector lies in $\operatorname{Ran} P$ and the leakage is identically zero. Otherwise, let $E_0^\gamma = \lambda_{\min}(H_{\text{prob}}|_{\operatorname{Ran}\Pi_\gamma})$ be the lowest problem-Hamiltonian energy in the sector and define the compressed invalid-state separation

$$\Gamma_\gamma = \lambda_{\min}\left(Q_\gamma H_{\text{prob}} Q_\gamma|_{\operatorname{Ran} Q_\gamma}\right) - E_0^\gamma. \tag{39}$$

If $\Gamma_\gamma > 0$, every normalized sector ground state ψ_γ satisfies

$$1 - \langle \psi_\gamma | P | \psi_\gamma \rangle \le \frac{\|B_\gamma\|^2}{\Gamma_\gamma^2 + \|B_\gamma\|^2}. \tag{40}$$

Moreover,

$$\Gamma_\gamma \ge m - \kappa - \eta_\gamma, \tag{41}$$

so the right-hand side of Eq. (40) remains valid with Γ_γ replaced by $g_\gamma = m - \kappa - \eta_\gamma > 0$.

Physically, the certificate compares the energetic separation Γ_γ protecting the feasible sector with the coupling $\|B_\gamma\|$ that leaks amplitude into invalid states. The kinetic floor η_γ is required because a sector-projected cover is a superposition and need not have zero hopping expectation. We do not claim that it changes the numerical certificate on every instance. The full proof and the cyclic-space caveat are in Appendix A.

This statement certifies the solution content of the lowest-energy state within a sector. It does not say that the ground vector has exact support on $\operatorname{Ran} P$: for finite penalties Eq. (40) controls, rather than eliminates, leakage. Exact support occurs only in an infinite penalty limit or in special decoupled cases.

The proposition applies to genuine commuting symmetry sectors. A cyclic projector need not commute with P, so cyclic-space solution probability is evaluated directly or through a separately constructed commuting projector.

3.1 Protocol-dependent gaps and initial-state dependence

For an invariant space $\mathcal{L}$, let $P_0^{\mathcal{L}}(s)$ be the ground projector of $H(s)|_{\mathcal{L}}$ and $g_0^{\mathcal{L}}(s) = \operatorname{rank} P_0^{\mathcal{L}}(s)$. Write the eigenvalues of the restriction, ordered with multiplicity, as $E_0^{\mathcal{L}}(s) \le E_1^{\mathcal{L}}(s) \le \cdots$. Whenever the indicated complementary level exists, we keep separate

$$\delta_{\mathrm{mult}}^{\mathcal{L}}(s) = E_1^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s), \tag{42}$$
$$\Delta_{\mathrm{exc}}^{\mathcal{L}}(s) = E_{g_0^{\mathcal{L}}(s)}^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s). \tag{43}$$

If $\dim \mathcal{L} = 1$, we set $\delta_{\mathrm{mult}}^{\mathcal{L}} = +\infty$; if $g_0^{\mathcal{L}} = \dim \mathcal{L}$, we set $\Delta_{\mathrm{exc}}^{\mathcal{L}} = +\infty$. These conventions express that the corresponding complementary level does not exist. The first vanishes whenever the ground state is degenerate; the second is the excitation gap above the whole manifold. A Schur multiplicity or an exact coincidence between irreps is therefore not called a gap closure when Eq. (43)'s second quantity stays positive.

Let E_0, E_0^{sym}, and $E_0^{\mathcal{K}}$ be the ground energies of $H(s)$ on $\mathcal{H}_{\mathrm{addr}}$, $\mathcal{H}_{\mathrm{sym}}$, and $\mathcal{K}_u$, respectively, at a common value of s. The nested spaces in Eq. (18) define the accessibility costs

$$\delta_{\mathrm{sym}} = E_0^{\mathrm{sym}} - E_0, \qquad \delta_{\mathrm{cyc}} = E_0^{\mathcal{K}} - E_0^{\mathrm{sym}}, \tag{44}$$
$$\delta_{\mathrm{acc}} = E_0^{\mathcal{K}} - E_0 = \delta_{\mathrm{sym}} + \delta_{\mathrm{cyc}}. \tag{45}$$

A positive δ_{acc} says that the protocol follows a globally excited branch. That branch can nevertheless have high cover probability.

For adiabatic evolution, the relevant object is an isolated spectral band, not merely the first distinct final eigenvalue [14]. Let $P_{\mathrm{track}}(s)$ be a continuous isolated spectral projector on $\mathcal{K}_u$ whose range contains the initial ground state. Set $H_{\mathcal{K}_u}(s) = H(s)|_{\mathcal{K}_u}$ and let $H_{\mathrm{track}}(s)$ be the compression of $H_{\mathcal{K}_u}(s)$ to $\operatorname{Ran} P_{\mathrm{track}}(s)$. Its complementary compression is

$$H_{\perp}(s) = H_{\mathcal{K}_u}(s)\big|_{\operatorname{Ran}(I_{\mathcal{K}_u} - P_{\mathrm{track}}(s))}, \tag{46}$$

where $I_{\mathcal{K}_u}$ is the identity on $\mathcal{K}_u$. For finite real sets A, B, write $\operatorname{dist}(A,B) = \min_{a\in A, b\in B} |a-b|$ and adopt the convention $\operatorname{dist}(A, \varnothing) = +\infty$. Define

$$\Delta_{\mathrm{ad}}(s) = \operatorname{dist}(\operatorname{spec} H_{\mathrm{track}}(s), \operatorname{spec} H_{\perp}(s)). \tag{47}$$

This projector-based definition retains multiplicity: if a complementary eigenvector has the same numerical eigenvalue as the tracked band, then $\Delta_{\mathrm{ad}} = 0$. An isolated continuous projector has constant rank. If its intended rank changes, an isolation gap has closed and a nondegenerate adiabatic formula cannot simply be continued through that point.

The four quantities have the following physical meanings:

quantity	physical meaning
δ_{mult}	splitting inside the lowest manifold; zero for a degenerate ground energy
Δ_{exc}	excitation gap above the entire lowest manifold
δ_{acc}	energy cost of confinement to the protocol-cyclic space
Δ_{ad}	isolation of the spectral band followed inside that space

The restriction is conditional on the protocol. If $[H(t), \Pi_\gamma] = 0$ and the initial state lies in $\operatorname{Ran} \Pi_\gamma$, its evolution remains there, but an initial state adapted to another irrep or a symmetry-breaking generator changes the accessible spectrum. This statement does not forbid transient concentration in a computationally marked subspace: it forbids only transition amplitude into symmetry sectors orthogonal to the one selected by the initial state.

4 Dark crossings in exact symmetry quotients

If Π_a and Π_b project onto inequivalent irreps of a symmetry preserved by the entire path, then

$$\Pi_a\, \partial_s H(s)\, \Pi_b = 0. \tag{48}$$

An equality of the two branch energies is therefore a global crossing but not an avoided crossing seen by a state confined to either block.

Two distinct mechanisms must be separated. First, the chosen retained-walk interpolation has an exactly diagonal point at which register-permutation multiplicities appear simultaneously. Second, an isolated pair of transverse branches can cross between inequivalent irreps. Only the latter mechanism is covered by the stability theorem below.

4.1 Orbit-resolved spectrum at hopping cancellation

The identities in Eq. (91) give

$$H(s) = a(s)I + c(s)M + sV, \tag{49}$$
$$a(s) = k\left(1 - s - \frac{s}{n_B - 1}\right), \tag{50}$$

where $c(s)$ is defined in Eq. (34). Its unique zero is

$$s_c = \frac{n_B - 1}{2n_B - 1}. \tag{51}$$

Theorem 6 (Orbit-resolved cancellation spectrum)**.** *At $s = s_c$,*

$$H(s_c) = s_c(kI + V). \tag{52}$$

Let $\varphi_1 < \cdots < \varphi_p$ be the distinct orbit-potential values. Then the distinct spectra on $\mathcal{H}_{\mathrm{addr}}$, $\mathcal{H}_{\mathrm{sym}}$, and $\mathcal{K}_u$ are all

$$\{s_c(k + \varphi_j) : 1 \le j \le p\}. \tag{53}$$

In particular, their ground energies and excitation gaps agree at s_c; if $p > 1$,

$$\Delta_{\mathrm{exc}}(s_c) = s_c(\varphi_2 - \varphi_1). \tag{54}$$

For $p = 1$, we adopt the convention $\Delta_{\mathrm{exc}}(s_c) = +\infty$.

Assume that at least one exact-k cover with distinct bases exists. Let N_F be the number of such base subsets and let N_{orb} be the number of their G_B-orbits. Then $\varphi_1 = 0$ and

$$g_0^{\mathrm{addr}}(s_c) = k!N_F, \qquad g_0^{\mathrm{sym}}(s_c) = N_{\mathrm{orb}}, \tag{55}$$

$$1 \le g_0^{\mathcal{K}}(s_c) \le N_{\mathrm{orb}}. \tag{56}$$

When $k = k^\star$, these are minimum-cover counts; otherwise they count feasible exact-k covers.

At cancellation, and only there along this path, the potential separation between orbit sites is exactly a spectral separation. For $s < s_c$ the full Hamiltonian is an irreducible stoquastic matrix [20] and has a unique ground state, whereas Eq. (55) is degenerate for every feasible instance with $k \ge 2$. The resulting global multiplicity closure contains nontrivial S_k components that are dark to the uniform state. The symmetric ground space can itself be degenerate when $N_{\mathrm{orb}} > 1$, so only a cover-transitive instance ($N_{\mathrm{orb}} = 1$) has a unique fixed-sector solution orbit.

The interior cancellation is a feature of the endpoint choice $H_{\mathrm{prob}} = H_{\mathrm{walk}} + V$. We retain the walk because this endpoint is a cost-dressed continuous-time search Hamiltonian rather than a purely classical cost operator. For the conventional diagonal endpoint V, the path $(1-s)H_{\mathrm{init}} + sV$ has hopping coefficient $-(1-s)/n_B$, which vanishes only at $s = 1$. The orbit-resolved diagonal spectrum and its permutation multiplicities still occur at that endpoint, but there is no interior cancellation point. We therefore do not present the closure as a path-independent property of adiabatic Set Cover.

4.2 Transverse dark crossings

Unlike the simultaneous diagonal closure, a transverse inter-sector crossing is locally stable within the symmetry-preserving class. We use standard differentiable perturbation theory for isolated Hermitian branches [21].

Theorem 7 (Stability of a transverse dark crossing)**.** *Let $H_\varepsilon(s) = H(s) + \varepsilon R(s)$ be a finite-dimensional Hermitian family that is continuously differentiable jointly in (s, ε), where $\varepsilon \in \mathbb{R}$ is the perturbation strength and $R(s)$ is Hermitian. Let Π_a, Π_b project onto inequivalent symmetry sectors. Suppose every $H_\varepsilon(s)$ commutes with both projectors. Assume that near $(s_\star, 0)$ the two restricted blocks have isolated differentiable reduced eigenvalue branches $E_a(s,\varepsilon)$ and $E_b(s,\varepsilon)$, simple on their multiplicity spaces but possibly carrying symmetry-enforced Schur multiplicity, such that*

$$E_a(s_\star, 0) = E_b(s_\star, 0), \qquad \partial_s(E_a - E_b)(s_\star, 0) \ne 0. \tag{57}$$

Then, for all sufficiently small $|\varepsilon|$, there is a unique nearby $s_\star(\varepsilon) = s_\star + O(|\varepsilon|)$ at which the two branches cross exactly. If the a branch is separated from the rest of its block by at least $\Delta_\star > 0$ throughout a neighbourhood of $s_\star$ at $\varepsilon = 0$, then its isolation gap remains at least $\Delta_\star/2$ whenever $|\varepsilon| \sup_{s\in[0,1]} \|R(s)\| \le \Delta_\star/4$ and the perturbed crossing stays in that neighbourhood.

Physically, the preserved symmetry forbids the two branches from hybridizing. A symmetry-preserving perturbation may move the crossing, but cannot open an avoided crossing between the sectors. The complete stability argument is in Appendix C.1.

4.3 A finite joint-symmetry audit

For the frozen eight-base incidence instance generated from a 2×4 vertex grid and specified explicitly in Appendix C.2, take $k = 3$, $\lambda = 5$, and $\mu = 10$. Its effective incidence group is $G_B \simeq D_4$, where D_4 is the order-eight dihedral group, and the $S_3 \times D_4$ fixed quotient has dimension 22, compared with 512 in the ordered address space. Cyclic closure from $|u\rangle$ also has dimension 22, as determined independently by the generator algebra. The uncatalysed path has a fixed-space rank closure at $s = 7/15$, showing that joint symmetry alone does not guarantee a rank-one accessible gap.

As a finite illustration of a dark crossing away from the cancellation point, use the single-register walk W from Eq. (5) and define

$$C_2 = -\sum_{1 \le r < t \le k} W^{(r)} W^{(t)}, \tag{58}$$

and use the endpoint-preserving path

$$H_\chi(s) = H(s) + 4\chi s(1-s)C_2, \tag{59}$$

where $\chi \in \mathbb{R}$ is the dimensionless catalyst strength. The operator C_2 commutes with both S_k and G_B, and its envelope vanishes at $s = 0, 1$. At $\chi = 2$, a crossing between different D_4 irreps occurs at $s \simeq 0.806454$, while the D_4-trivial gap there is approximately 0.258. The reduced-branch slope difference is approximately -4.54, so the crossing is transverse. This diagnostic is not used in the asymptotic theorem. Appendix C.2 specifies its numerical protocol, minimum-gap scan, and residuals.

4.4 Even cycles as a cover-transitive illustration

Express minimum vertex cover on the even cycle C_n, $n = 2k$, as set cover: bases are vertices, tasks are edges, and each base covers its two incident edges.

The only minimum covers are the two alternating subsets. They are exchanged by a one-site rotation and hence form one orbit of the dihedral group D_n. Thus this family is cover-transitive, with $N_F = 2$ and $N_{\mathrm{orb}} = 1$.

Assume

$$\mu \geq \lambda > 0. \tag{60}$$

Any repeated k-tuple uses fewer than $k = k^\star$ distinct bases and therefore leaves at least one edge uncovered; its penalty is at least $\lambda + \mu$. A distinct noncover has penalty at least λ, and this value is attained by

$$S_\star = \{1\} \cup \{2j : 1 \leq j \leq k-1\}, \tag{61}$$

which leaves exactly one edge uncovered. Theorem 6 therefore gives

$$s_c = \frac{n-1}{2n-1}, \qquad g_0^{\mathrm{addr}}(s_c) = 2 \cdot k!, \tag{62}$$

$$g_0^{\mathrm{sym}}(s_c) = g_0^{\mathcal{K}}(s_c) = 1, \quad \Delta_{\mathrm{exc}}^{\mathcal{H}_{\mathrm{sym}}}(s_c) = \lambda \frac{n-1}{2n-1}. \tag{63}$$

The fixed-sector excitation gap therefore increases monotonically from $3\lambda/7$ at $n = 4$ towards $\lambda/2$ as $n \to \infty$, without attaining that limit, while the global ground multiplicity becomes $2 \cdot k!$. This is a diagonal, permutation-induced multiplicity closure, not a transverse two-branch crossing; Theorem 7 does not assert its stability under a general symmetry-preserving perturbation.

Equation (63) is pointwise and does not bound the minimum gap along the retained-walk linear interpolation. The next section studies a different parent path on the same family. Its endpoint, initial state, and generator algebra are distinct from those above.

5 A separate parent path and its accessible gap

We now change protocols while retaining the same even-cycle covering instances. The parent Hamiltonians below do not interpolate between H_{init} and H_{prob}: their state space, initial state, endpoint, and generator algebra are specified independently. This second construction controls an entire accessible path rather than the single cancellation point of Section 4. Its purpose is a gap certificate, not a claim that vertex cover on a cycle is classically difficult.

5.1 Even-cycle set cover and its joint symmetry

Let the bases be the vertices of the even cycle C_n, $n = 2k$, and let the tasks be its edges. Base b covers the two edges incident on b. A k-subset $S \subset \mathbb{Z}_n$, where $\mathbb{Z}_n$ denotes the vertex labels modulo n, is therefore a cover precisely when it is one of the two alternating subsets. The two solutions are exchanged by a one-site rotation and form one orbit of the dihedral incidence automorphism group D_n, generated by cycle rotations and reflections.

We work in the injective, register-symmetric subspace, with basis $\{|S\rangle : |S| = k\}$. Equivalently, this is the symmetric Dicke embedding of the hard-core sector of the original registers. Define

$$U(S) = |\{i \in \mathbb{Z}_n : i \notin S,\ i+1 \notin S\}|. \tag{64}$$

At half filling, $U(S)$ is also the number of adjacent occupied pairs. Thus $U = 0$ exactly on the two minimum covers.

For a dimensionless annealing parameter $a \geq 0$, consider the continuous-time Metropolis generator with a Johnson-graph proposal. The Johnson graph $J(n,k)$ has the k-subsets of $[n]$ as vertices and joins S to T exactly when $|S \triangle T| = 2$, i.e., one selected vertex is replaced. For distinct adjacent S, T, the transition rate is

$$q_a(S,T) = \frac{1}{n} \min\{1, n^{-a[U(T)-U(S)]}\}, \qquad |S \triangle T| = 2, \tag{65}$$

and $q_a(S,T) = 0$ otherwise. Its reversible measure is

$$\pi_a(S) = Z_a^{-1} n^{-aU(S)}. \tag{66}$$

Here $Z_a = \sum_{|S|=k} n^{-aU(S)}$ normalizes the probability distribution. The symmetric-discriminant construction is standard for reversible Markov chains and Gibbs-amplitude ground states [10, 11]. Here H_a has diagonal entry $\langle S|H_a|S\rangle = \sum_{T \neq S} q_a(S,T)$ and, for $|S \triangle T| = 2$, off-diagonal entries

$$\langle T|H_a|S\rangle = -\sqrt{q_a(S,T) q_a(T,S)} \tag{67}$$

$$= -\frac{1}{n} n^{-a|U(T)-U(S)|/2}. \tag{68}$$

All remaining off-diagonal entries are zero, so these equations define H_a completely. It is frustration free and has the known ground state

$$|\psi_a\rangle = Z_a^{-1/2} \sum_{|S|=k} n^{-aU(S)/2} |S\rangle. \tag{69}$$

At $a = 0$, this is the Dicke state $|J_{n,k}\rangle = \binom{n}{k}^{-1/2} \sum_{|S|=k} |S\rangle$ [22], and the parent Hamiltonian $H_{a=0}$ is exactly the Johnson-graph Laplacian at rate $1/n$. It should not be confused with the restriction of H_{init} from Eq. (8), which agrees with a Johnson Laplacian only up to an additive constant after projection to the injective subspace. The whole parent path commutes with D_n.

This protocol has its own generator algebra and cyclic space, distinct from Eq. (17):

$$\mathfrak{A}_{\mathrm{par}} = \mathrm{alg}^*(\{H_a : 0 \leq a \leq 7\} \cup \{I\}), \tag{70}$$

$$\mathcal{K}_J^{\mathrm{par}} = \mathfrak{A}_{\mathrm{par}} |J_{n,k}\rangle. \tag{71}$$

Here I is the identity on the half-filled hard-core space. Let $\mathcal{H}^{D_n}$ denote the D_n-fixed subspace of the half-filled hard-core space. Define the fixed-space and cyclic gaps by

$$\Delta_{D_n}(a) = \mathrm{gap}(H_a|_{\mathcal{H}^{D_n}})\,, \tag{72}$$

$$\Delta^{\mathrm{cyc}}_{\mathrm{par}}(a) = \mathrm{gap}\Big(H_a|_{\mathcal{K}^{\mathrm{par}}_J}\Big)\,. \tag{73}$$

Proposition 8 (Cyclic accessibility of the parent path)**.** *For every even $n = 2k \geq 6$ and $0 \leq a \leq 7$,*

1. *$\mathcal{K}^{\mathrm{par}}_J$ is a common reducing subspace for the family $\{H_a\}_{0\leq a\leq 7}$ and $\mathcal{K}^{\mathrm{par}}_J \subseteq \mathcal{H}^{D_n}$;*
2. *the unique ground state $|\psi_a\rangle$ belongs to $\mathcal{K}^{\mathrm{par}}_J$; and*
3. $$\Delta^{\mathrm{cyc}}_{\mathrm{par}}(a) \geq \Delta_{D_n}(a). \tag{74}$$

Thus a lower bound proved in the D_n-fixed space is also a lower-bound certificate for the dynamically accessible gap of this parent protocol.

Physically, the protocol explores no more than the D_n-fixed sector, and restricting further to its true cyclic space cannot introduce an excitation below the fixed-sector gap when the unique ground state is shared. The full argument is in Appendix D.1.

The number of configurations at energy u is

$$N_u = \frac{n}{k-u}\binom{k-1}{u}^2, \qquad 0 \leq u < k. \tag{75}$$

At $a_f = 7$, this shell count implies

$$1 - \langle\psi_{a_f}|\Pi_{U=0}|\psi_{a_f}\rangle = O(n^{-5}). \tag{76}$$

Here $\Pi_{U=0}$ is the orthogonal projector onto the two-dimensional span of the configurations satisfying $U(S) = 0$. Appendix D.2 gives the run decomposition and ratio bound.

5.2 A uniform accessible-gap theorem

We first separate the cycle geometry from the zero-range kinetics.

Proposition 9 (Zero-range reduction)**.** *For every even $n = 2k \geq 6$ and every $a \geq 0$, the gap of H_a in the D_n-fixed space obeys*

$$\Delta_{D_n}(a) \geq \frac{2}{n}[1-\cos(2\pi/k)]\,\mathrm{gap}\,\mathrm{ZRP}(K_k, r_a, k), \tag{77}$$

where $r_a(0) = 0$, $r_a(1) = n^{-a}$, $r_a(j) = 1$ for $j \geq 2$, and K_k is the uniform mean-field kernel $K_k(i,j) = 1/k$ on k boxes. The notation $\mathrm{ZRP}(K_k, r_a, k)$ denotes the zero-range process with this jump kernel, rate function r_a, and k particles; its gap is the Poincaré constant of the corresponding generator. Consequently, any uniform polynomial lower bound for this mean-field gap gives a uniform polynomial accessible-gap certificate for the parent path.

The factors in Eq. (77) have distinct physical origins: $2/n$ is the update-rate normalization, $1 - \cos(2\pi/k)$ is the long-wavelength transport cost of the cycle, and the zero-range gap measures the remaining occupation-redistribution bottleneck. The exact isometry, rate comparison, and normalization are proved in Appendix E.

Proposition 10 (Uniform accessible gap)**.** *For every even $n = 2k \geq 6$ and every $a \geq 0$,*

$$\mathrm{gap}\,\mathrm{ZRP}(K_k, r_a, k) \geq 2n^{-a}k^{-3}, \tag{78}$$

$$\Delta_{D_n}(a) \geq 1024\, n^{-(a+6)}. \tag{79}$$

Consequently, along $0 \leq a \leq 7$, $\min_a \Delta_{D_n}(a) \geq 1024\, n^{-13}$.

Physically, the polynomial bound accumulates three conservative costs: normalized subset updates, long-wavelength transport around the cycle, and redistribution of vacancies between effective boxes. Its exponent is a certificate, not a prediction of the finite-size gap scaling. The complete comparison is in Appendix E.

The gap theorem can be converted into an adiabatic statement in the abstract Hamiltonian-evolution model. This does not by itself supply a gate-level implementation of the path.

Corollary 11 (Conditional adiabatic preparation)**.** *In the abstract Hamiltonian-access model, let $a(s) = 7s$, $0 \leq s \leq 1$, and evolve from $|J_{n,k}\rangle$ under the restricted path $H_{a(s)}|_{\mathcal{K}^{\mathrm{par}}_J}$ for total time T. There is a universal constant $C > 0$ such that, for every $0 < \epsilon < 1$, the sufficient condition*

$$T \geq C\epsilon^{-1}n^{41}(\log n)^2 \tag{80}$$

produces a final state at distance $O(\epsilon)$ from $|\psi_7\rangle$, up to phase. Its cover probability is therefore $1 - O(n^{-5}) - O(\epsilon)$. This runtime excludes the costs of preparing the Dicke state and realizing or simulating $H_{a(s)}$.

Physically, only the isolation gap inside $\mathcal{K}^{\mathrm{par}}_J$ enters the adiabatic theorem; spectral branches in orthogonal symmetry sectors do not set the preparation time of this protocol. The large runtime exponent instead reflects deliberately conservative derivative and inverse-gap bounds and does not include Hamiltonian-simulation or state-preparation costs. The complete adiabatic and measurement estimates are in Appendix D.3.

Exact D_n-orbit quotients for $n = 6, 8, \ldots, 18$ show finite-size gaps much larger than the conservative theorem, but are not used to infer scaling. At $n = 6$ and $a = 7$, for example, the certificate is $1024\, 6^{-13} = 7.84\times 10^{-8}$ whereas the reconstructed gap is 0.166670, a factor 2.1×10^6 larger. The analytical bottleneck is the uniform mean-field estimate $2n^{-a}k^{-3}$; the additional $2/n$ update normalization and cycle transport factor are displayed separately in Eq. (77). The grid and all intermediate values are reported in Appendix F and the archived data.

The two constructions share only the instance family, the cover-measurement objective, and the orbital language. No dynamical interpolation or complexity transfer between them is assumed.

6 Finite-instance cyclic audit

Figure 2 isolates the two strongest finite diagnostics. They show separately that cyclic exclusion can occur inside the trivial symmetry sector and that the cancellation point can amplify the global ground rank without changing the rank followed by the protocol. All finite-instance Hamiltonians reported here and in Appendix F use the penalty energies $\lambda = 5$ and $\mu = 10$ in Eq. (7).

At the problem endpoint H_{prob}, write Δ_{full}, Δ_{sym}, and Δ_{cyc} for the excitation gaps on $\mathcal{H}_{\mathrm{addr}}$, $\mathcal{H}_{\mathrm{sym}}$, and $\mathcal{K}_u$. The energy costs δ_{sym} and δ_{cyc} are those in Eq. (45). If $\mathcal{G}_{\mathrm{cyc}}$ is the cyclic ground eigenspace, define its worst-case cover probability by

$$p_{\mathrm{opt}}^{\min} = \min_{\substack{\psi \in \mathcal{G}_{\mathrm{cyc}} \\ \|\psi\|=1}} \langle \psi | P | \psi \rangle, \tag{81}$$

where P is the feasible-cover projector from Section 3.

For `rnd-b8-t7-p50-s002`, the endpoint calculation gives

$$\begin{aligned} |E_0^{\mathcal{K}} - E_1^{\mathrm{sym}}| &< 10^{-14}, \\ \big| |\langle E_1^{\mathrm{sym}} | E_0^{\mathcal{K}} \rangle|^2 - 1 \big| &< 10^{-14}. \end{aligned} \tag{82}$$

The symmetric ground-vector weight in $\mathcal{K}_u$ is below 10^{-26}. Because G_B is trivial here, both symmetric levels carry the trivial S_3 irrep [3]: the exclusion is caused by cyclic closure, not by a further symmetry sector. The 19 missing directions may be chosen shell-local by Eq. (32); their deficits at $\Phi = 0, 5, 15, 20, 25$ are respectively $2, 7, 4, 4, 2$.

For `grid-2x4`, the global ground rank is one for $s < s_c$, becomes $3! \times 12 = 72$ at $s_c = 7/15$, and is two at the problem endpoint. This $1 \to 72 \to 2$ comparison collects the rank information that otherwise sits in separate path and endpoint tables.

Across all eleven frozen instances, five have $\mathcal{K}_u \subsetneq \mathcal{H}_{\mathrm{sym}}$, but only one has $\delta_{\mathrm{cyc}}(1) > 0$; eight have $\delta_{\mathrm{sym}}(1) > 0$ and nine have $\delta_{\mathrm{acc}}(1) > 0$. On the audited path grid, $|\delta_{\mathrm{cyc}}(s)| \le 2 \times 10^{-14}$ for every $s \le s_c$, as predicted by Theorem 4. The only positive endpoint cyclic cost occurs above s_c, where the hopping coefficient has changed sign and the Perron–Frobenius argument no longer applies. Every cyclic endpoint ground projector has rank one, with $p_{\mathrm{opt}}^{\min} \ge 0.9944$. These are finite diagnostics, not scaling evidence. Appendix F gives the full tables, shell-closure conditioning, residual-based degeneracy rule, irrep classification, and OR-Library provenance.

7 Discussion and conclusion

For a fixed initial state and a symmetry-preserving interpolation, the physically relevant spectrum is neither the full spectrum nor, in general, the entire group-fixed spectrum. It is the spectrum of the Hamiltonian restricted to the cyclic space generated by that protocol. Thus $\mathcal{K}_u \subseteq \mathcal{H}_{\mathrm{sym}}$ is the central distinction: changing the initial state, the generated algebra, or a preserved symmetry can change the accessible spectrum and therefore the applicable gap. The resulting inaccessibility statement is correspondingly conditional on state preparation and on every symmetry preserved along the path.

Within this framework, orbit invariance identifies unlabelled candidate covers as the natural sites of the fixed-sector quotient, while the shell closure determines which amplitude patterns among those sites the endpoint algebra can generate. The sector-resolved localization proposition certifies cover-measurement probability with the required kinetic floor. Exact orbit quotients separate ground multiplicity, excitation gaps, isolation gaps, and accessibility costs, and expose coincidences between irreducible sectors that are dark to the symmetric protocol. At the hopping-cancellation point, the retained-walk interpolation has an exact global multiplicity closure for every feasible fixed-cardinality instance. The even cycle is a cover-transitive illustration whose fixed excitation gap is at least $3\lambda/7$ and approaches $\lambda/2$ from below at that point. A distinct Johnson–Metropolis parent path has a uniformly polynomial cyclic gap bounded below by $1024\, n^{-13}$ along the path, and consequently a conditional polynomial adiabatic runtime in the abstract Hamiltonian-access model.

These results establish a structural separation, not a practical quantum speedup. The cycle instances are classically solvable, Dicke-state preparation has a cost that must be counted, and the Gibbs parent endpoint is not the retained-walk problem Hamiltonian H_{prob}. Likewise, the address count $k\lceil \log_2 n_B \rceil$ is not a complete hardware resource estimate: incidence access, flags, ancillas, and gate depth require a separate implementation analysis.

A natural next step is a decorated family whose classical cover structure is nontrivial while its joint orbit quotient, dark global branch, and polynomial accessible gap remain stable under controlled symmetry-preserving perturbations. More broadly, spectral claims for symmetric Hamiltonian algorithms should identify not only a Hamiltonian and its global gap, but also the initial state, preserved symmetry, generated cyclic space, and isolated band actually followed by the protocol.

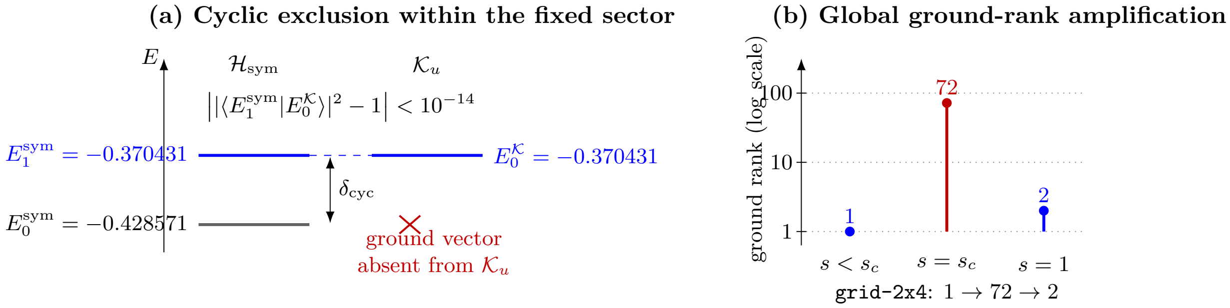

Figure 2: Two finite diagnostics of cyclic accessibility. (a) At the problem endpoint of `rnd-b8-t7-p50-s002`, the symmetric ground vector has negligible projection onto $\mathcal{K}_u$, while the cyclic ground state coincides, within residual-based precision, with the first symmetric excitation. (b) For `grid-2x4`, irreducible stoquasticity gives global ground rank one for $s<s_c$, hopping cancellation raises it to $3!\times 12=72$ at $s_c=7/15$, and the endpoint rank is two. Panel (b) uses a logarithmic rank axis.

Acknowledgements

This study was financed, in part, by the São Paulo Research Foundation (FAPESP), Brazil, grant 2026/04387-7. The author thanks Marcos César de Oliveira for supervision and discussions throughout this project.

Author contributions

Fabrício de Souza Luiz conceived the project, developed the mathematical analysis, implemented and validated the numerical calculations, and wrote the manuscript. Large-language-model tools from OpenAI were used to assist with editorial reorganization, language revision, bibliographic discovery, LaTeX editing, and the development and checking of reproducibility code. The author reviewed all model-assisted material, verified the mathematical and numerical results, and takes full responsibility for the content of the work.

Code and data availability

The frozen instance registry, source code, residual-based numerical checks, and machine-readable tables accompanying this preprint are available in the versioned repository `https://github.com/fsluiz/manuscript-encoding-symmetry-limits`. A version-specific Zenodo DOI will be added before journal submission. Original software is released under the MIT License; original data and documentation are released under CC BY 4.0. Third-party materials, including OR-Library incidence data and the journal class file, retain their source licenses and provenance.

References

[1] Richard M. Karp. "Reducibility among combinatorial problems". In R. E. Miller, J. W. Thatcher, and J. D. Bohlinger, editors, Complexity of Computer Computations. Pages 85–103. Plenum, New York (1972).

[2] Andrew Lucas. "Ising formulations of many NP problems". Frontiers in Physics **2**, 5 (2014).

[3] Stuart Hadfield, Zhihui Wang, Bryan O'Gorman, Eleanor G. Rieffel, Davide Venturelli, and Rupak Biswas. "From the quantum approximate optimization algorithm to a quantum alternating operator ansatz". Algorithms **12**, 34 (2019).

[4] Benjamin Tan, Marc-Antoine Lemonde, Supanut Thanasilp, Jirawat Tangpanitanon, and Dimitris G. Angelakis. "Qubit-efficient encoding schemes for binary optimisation problems". Quantum **5**, 454 (2021). arXiv:2007.01774.

[5] Adam Glos, Aleksandra Krawiec, and Zoltán Zimborás. "Space-efficient binary optimization for variational quantum computing". npj Quantum Information **8**, 39 (2022). arXiv:2009.07309.

[6] Nicholas Chancellor. "Domain wall encoding of discrete variables for quantum annealing and QAOA". Quantum Science and Technology **4**, 045004 (2019). arXiv:1903.05068.

[7] Ruslan Shaydulin, Stuart Hadfield, Tad Hogg, and Ilya Safro. "Classical symmetries and the quantum approximate optimization algorithm". Quantum Information Processing **20**, 359 (2021). arXiv:2012.04713.

[8] Hari Krovi and Todd A. Brun. "Quantum walks on quotient graphs". Physical Review A **75**, 062332 (2007). arXiv:quant-ph/0701173.

[9] Yousef Saad. "Analysis of some Krylov subspace approximations to the matrix exponential operator". SIAM Journal on Numerical Analysis **29**, 209–228 (1992).

[10] Mario Szegedy. "Quantum speed-up of Markov chain based algorithms". In Proc. 45th Annual IEEE Symposium on Foundations of Computer Science (FOCS). Pages 32–41. (2004).

[11] Rolando D. Somma, Sergio Boixo, Howard Barnum, and Emanuel Knill. "Quantum simulations of classical annealing processes".

Physical Review Letters **101**, 130504 (2008). arXiv:0804.1571.

[12] Edward Farhi, Jeffrey Goldstone, Sam Gutmann, and Michael Sipser. “Quantum computation by adiabatic evolution” (2000). arXiv:quant-ph/0001106.

[13] Tameem Albash and Daniel A. Lidar. “Adiabatic quantum computation”. Reviews of Modern Physics **90**, 015002 (2018). arXiv:1611.04471.

[14] Sabine Jansen, Mary-Beth Ruskai, and Ruedi Seiler. “Bounds for the adiabatic approximation with applications to quantum computation”. Journal of Mathematical Physics **48**, 102111 (2007). arXiv:quant-ph/0603175.

[15] Jean-Pierre Serre. “Linear representations of finite groups”. Volume 42 of Graduate Texts in Mathematics. Springer. New York (1977).

[16] William Burnside. “Theory of groups of finite order”. Cambridge University Press. Cambridge (1911). 2 edition. url: https://www.cambridge.org/core/books/theory-of-groups-of-finite-order/D3734FEB30904C706780248BE3A9EF20.

[17] J. Howard Redfield. “The theory of group-reduced distributions”. American Journal of Mathematics **49**, 433–455 (1927).

[18] George Pólya. “Kombinatorische anzahlbestimmungen für gruppen, graphen und chemische verbindungen”. Acta Mathematica **68**, 145–254 (1937).

[19] Roger A. Horn and Charles R. Johnson. “Matrix analysis”. Cambridge University Press. Cambridge (2012). 2 edition.

[20] Sergey Bravyi, David P. DiVincenzo, Roberto I. Oliveira, and Barbara M. Terhal. “The complexity of stoquastic local hamiltonian problems”. Quantum Information and Computation **8**, 361–385 (2008). arXiv:quant-ph/0606140.

[21] Tosio Kato. “Perturbation theory for linear operators”. Classics in Mathematics. Springer. Berlin (1995).

[22] Andreas Bärtschi and Stephan Eidenbenz. “Deterministic preparation of Dicke states”. In Fundamentals of Computation Theory. Volume 11651 of Lecture Notes in Computer Science, pages 126–139. Springer (2019). arXiv:1904.07358.

[23] Jonathan Hermon and Justin Salez. “A version of Aldous’ spectral-gap conjecture for the zero range process”. The Annals of Applied Probability **29**, 2217–2229 (2019). arXiv:1808.00325.

[24] Bradley Efron. “Increasing properties of Pólya frequency functions”. The Annals of Mathematical Statistics **36**, 272–279 (1965).

[25] Pietro Caputo. “On the spectral gap of the Kac walk and other binary collision processes”. ALEA Latin American Journal of Probability and Mathematical Statistics **4**, 205–222 (2008). arXiv:0807.3415.

[26] J. E. Beasley. “Or-library: Distributing test problems by electronic mail”. Journal of the Operational Research Society **41**, 1069–1072 (1990).

[27] J. E. Beasley. “OR-Library”. http://people.brunel.ac.uk/~mastjjb/jeb/orlib/scpinfo.html (1990). Accessed 2026-05-14.

A Proof of sector localization

We prove Proposition 5 and record why its sector hypothesis cannot be silently replaced by cyclic restriction. The variational principle applied inside $\operatorname{Ran}\Pi_\gamma$ gives $E_0^\gamma \le \eta_\gamma$. Combining this with Eq. (36) yields

$$\Gamma_\gamma \ge m - \kappa - \eta_\gamma, \tag{83}$$

which is Eq. (41).

For a normalized sector ground vector, write $p = P_\gamma\psi_\gamma$ and $q = Q_\gamma\psi_\gamma$. Since P, Q, and Π_γ commute, these are orthogonal components and the lower block of the eigenvalue equation is

$$(Q_\gamma H_{\text{prob}} Q_\gamma - E_0^\gamma)q = -B_\gamma p. \tag{84}$$

If $Q_\gamma = 0$, then $q = 0$ identically. Otherwise positivity of Γ_γ makes the compressed operator on the left invertible with inverse norm at most Γ_γ^{-1}. Therefore

$$\|q\| \le \frac{\|B_\gamma\|}{\Gamma_\gamma}\|p\|. \tag{85}$$

Using $\|p\|^2 + \|q\|^2 = 1$ and solving for $\|q\|^2$ gives Eq. (40). Replacing Γ_γ by the positive lower bound $m - \kappa - \eta_\gamma$ weakens the right-hand side and proves the stated certificate.

For finite penalties, the proposition controls rather than eliminates leakage; exact support on covers requires an infinite-penalty limit or a special decoupling. Moreover, if $\Pi_{\mathcal{K}}$ denotes the orthogonal projector onto $\mathcal{K}_u$, it need not commute with P, so $P\Pi_{\mathcal{K}}$ need not be an orthogonal projector. The proposition therefore applies directly to genuine commuting symmetry sectors. Cyclic-space solution probability must be evaluated directly or bounded with a separately constructed commuting projector.

B Joint representation and orbit quotient

The incidence automorphism group G_A acts on the address Hilbert space only through its base projection. If $\rho(\pi_B, \pi_T) = \pi_B$, then elements with $\pi_B = \mathrm{id}$ form the kernel, and the faithfully represented group is $G_B \simeq G_A/\ker\rho$. Since the S_k register action commutes with the G_B base action, complete reducibility gives

$$\mathcal{H}_{\text{addr}} \simeq \bigoplus_{\alpha,\beta} V_\alpha^{S_k} \otimes V_\beta^{G_B} \otimes \mathcal{M}_{\alpha\beta}. \tag{86}$$

Every invariant Hamiltonian lies in the commutant and hence has the form in Eq. (14). The identity factors generate Schur multiplicities; equality of eigenvalues belonging to different (α,β) blocks is not enforced by this statement.

B.1 Orbit invariance and enumeration

Register permutations leave the multiset $\{\{b_1,\dots,b_k\}\}$ unchanged, so they preserve both U and D. A base automorphism $g \in G_B$ is a bijection; hence $b_r = b_s$ if and only if $g(b_r) = g(b_s)$, proving invariance of D. For U, choose any task permutation π_T such that $(g,\pi_T) \in G_A$. If S is the underlying set of selected bases, write

$$\mathcal{U}(S) = \{t \in \mathcal{T} : A_{bt} = 0 \text{ for every } b \in S\}. \tag{87}$$

The incidence-automorphism identity gives $\mathcal{U}(gS) = \pi_T\mathcal{U}(S)$, so the two uncovered-task sets have the same cardinality. This proves Lemma 1. The cardinality is independent of which compatible π_T is chosen, as it must be because only the faithfully represented base action enters the Hilbert space.

For completeness, let $\mathcal{O}$ and $\mathcal{O}'$ be computational-basis orbits of the joint action and choose $x \in \mathcal{O}$. Invariance gives

$$\langle \mathcal{O}'|H|\mathcal{O}\rangle = \frac{1}{\sqrt{|\mathcal{O}||\mathcal{O}'|}} \sum_{x'\in\mathcal{O}} \sum_{y\in\mathcal{O}'} H_{yx'} \tag{88}$$

$$= \sqrt{\frac{|\mathcal{O}|}{|\mathcal{O}'|}} \sum_{y\in\mathcal{O}'} H_{yx}, \tag{89}$$

which proves Eq. (23). The orbit states are therefore an exact orthonormal basis of the fixed space, not an effective approximation.

After quotienting by register permutations, basis vectors are multisets of size k over the bases. If $g \in G_B$ has $c_\ell(g)$ cycles of length ℓ, the trace of g on the symmetric tensor power $\mathrm{Sym}^k(\mathbb{C}^{n_B})$ is the coefficient

$$[z^k]\prod_{\ell\ge 1}(1-z^\ell)^{-c_\ell(g)}. \tag{90}$$

The Cauchy–Frobenius orbit-counting lemma, equivalently the Pólya–Redfield cycle-index evaluation for multisets, averages this character over G_B and projects onto the trivial irrep. This gives Eq. (21). It counts $\mathcal{H}_{\text{sym}}$ only. Whether $|u\rangle$ is cyclic for the restricted endpoint algebra is a separate closure calculation.

B.2 Proof of the shell-resolved cyclic closure

The endpoint operators satisfy

$$H_{\text{init}} = kI - \frac{M}{n_B}, \qquad H_{\text{walk}} = \frac{M - kI}{n_B - 1}. \tag{91}$$

Thus $M = n_B(kI - H_{\text{init}})$ belongs to the endpoint algebra, H_{walk} follows from M, and $V = H_{\text{prob}} - H_{\text{walk}}$ also belongs to it. Conversely, Eq. (91) reconstructs both endpoints from M, V, I. This proves Lemma 3.

Because $\widehat{V} = \sum_j \varphi_j R_j$, invariance under $\widehat{V}$ is equivalent to invariance under every R_j: each R_j is a Lagrange interpolation polynomial in $\widehat{V}$. The initial shell vectors belong to the cyclic space since

$$\widehat{V}^i w = \sum_{j=1}^{p} \varphi_j^i R_j w, \qquad 0 \le i < p, \tag{92}$$

and the Vandermonde matrix of the distinct φ_j is invertible. Every $R_j w$ is nonzero by Eq. (28). The iteration in Eq. (30) therefore begins inside $\mathcal{K}_u$, remains there, and stabilises after finitely many steps. Its limit contains w and is invariant under every R_j and under $\widehat{M}$, hence under $\widehat{V}$ and the complete endpoint algebra. Minimality of the cyclic space proves equality. The p nonzero shell vectors are mutually orthogonal, which proves Eq. (33). If every shell is a singleton, those vectors are nonzero multiples of all orbit-basis vectors, proving the injectivity criterion and its contrapositive.

Since every R_j is an orthogonal projector in the endpoint algebra, $R_j\mathcal{K}_u \subseteq \mathcal{K}_u$. The mutually orthogonal projectors sum to the identity on the orbit space, so $\mathcal{K}_u = \bigoplus_j R_j\mathcal{K}_u$. Taking the orthogonal complement inside $\bigoplus_j \operatorname{Ran} R_j$ gives the second identity in Eq. (32). Thus a basis of inaccessible directions may always be chosen shell-local, although this statement alone does not assign an orientation or current interpretation to those directions.

It remains to prove Eq. (35). For s below the unique zero of $c(s)$, every off-diagonal entry of the computational-basis Hamiltonian is nonpositive. Equation (23) preserves that sign, and the orbit hopping graph is connected: a sequence of one-register substitutions connects any two size-k multisets. After a sufficiently large scalar shift, Perron–Frobenius [19] therefore gives a simple symmetric ground vector $\psi_0(s)$ with strictly positive orbit coordinates. Its overlap with w is nonzero. The rank-one spectral projector onto $\psi_0(s)$ is a polynomial in the finite-dimensional quotient Hamiltonian, which belongs to the algebra generated by $\widehat{M}$ and $\widehat{V}$. Applying that projector to w proves $\psi_0(s) \in \mathcal{K}_u$.

At the zero of $c(s)$, the Hamiltonian is a scalar plus a positive multiple of $\widehat{V}$. If $R_{\min}$ denotes its lowest nonempty potential shell, then $R_{\min}w \neq 0$ and belongs to $\mathcal{K}_u$. It is a ground vector, so the symmetric and cyclic ground energies again agree. Continuity also covers $s = 0$. This completes the proof of Theorem 4.

B.3 Proof of the cancellation theorem

Substitution of Eq. (91) into the linear path gives Eq. (50). Solving $c(s) = 0$ gives $s = s_c$, and direct evaluation gives $a(s_c) = ks_c$ and s_cV for the remaining potential term. This proves Eq. (52).

In the computational basis, V is diagonal with entries Φ; by Lemma 1, each potential shell is a union of joint orbits. Therefore every φ_j occurs in the full and fixed-space spectra. It also occurs in the cyclic restriction because the nonzero vector R_jw belongs to $\mathcal{K}_u$ and is an eigenvector of $H(s_c)$. No other eigenvalue is possible in any restriction, proving Eqs. (53) and (54). If there is no second shell, the excitation gap is assigned the convention $+\infty$.

For a feasible exact-k subset, all $k!$ orderings are distinct computational-basis ground vectors. Conversely, zero potential requires an exact-k cover with no repeated bases, so this accounts for the complete full ground rank. Register quotienting turns those tuples into subsets, and quotienting by G_B leaves one fixed-space vector per feasible-cover orbit. Finally, R_1w supplies at least one cyclic ground vector, while $\mathcal{K}_u \subseteq \mathcal{H}_{\text{sym}}$ gives the upper rank bound. This proves Eqs. (55) and (56), and hence Theorem 6.

C Stability and the finite D_4 diagnostic

C.1 Proof of the stability theorem

Simple isolated reduced eigenvalues of a differentiable Hermitian family define differentiable local branches; the identity factor on an irrep may replicate such a branch without changing its scalar energy. Apply the implicit-function theorem to $F(s,\varepsilon) = E_a(s,\varepsilon) - E_b(s,\varepsilon)$ and use Eq. (57). This gives a unique nearby root $s_\star(\varepsilon) = s_\star + O(|\varepsilon|)$. Commutation with the symmetry projectors makes every inter-sector matrix element vanish, so the equality cannot become an avoided crossing. Finally, Weyl's bound moves each eigenvalue of the a block by at most $|\varepsilon|\|R(s)\|$. An internal spectral distance can shrink by at most twice this quantity, which proves the isolation-gap statement in Theorem 7.

C.2 Catalyst and numerical audit

The frozen instance has bases and tasks labelled $0,\ldots,7$ and incidence matrix

$$A = \begin{pmatrix} 1&1&0&0&1&0&0&0\\ 1&1&1&0&0&1&0&0\\ 0&1&1&1&0&0&1&0\\ 0&0&1&1&0&0&0&1\\ 1&0&0&0&1&1&0&0\\ 0&1&0&0&1&1&1&0\\ 0&0&1&0&0&1&1&1\\ 0&0&0&1&0&0&1&1 \end{pmatrix}. \tag{93}$$

It is the closed-neighbourhood incidence relation of the 2×4 vertex grid. Because task labels may be permuted independently of base labels, its effective incidence group has order eight and is isomorphic to D_4; this need not equal the geometric automorphism group obtained by forcing the same permutation on both colours. For $k = 3$, the $S_3 \times D_4$ fixed quotient and the cyclic closure from $|u\rangle$ both have dimension 22, compared with 512 in the ordered address space. The latter equality is an output of the closure calculation, not an assumption based on symmetry. The orbit-isometry, endpoint-invariance, and cyclic-invariance residuals are below 10^{-13}.

For the catalyst in Eqs. (58)– (59), set $\chi = 2$. The remaining global crossing is at $s \simeq 0.806454$. The two branches belong to different D_4 irreps, Eq. (48) has residual 4.7×10^{-14}, the reduced-branch slope difference is approximately -4.54, and the D_4-trivial gap there is approximately 0.258. The nontrivial branch is a rank-two E multiplet; the derivative restricted to that multiplet is scalar to numerical precision. The trivial-sector gap was evaluated on a uniform 2001-point grid and locally minimized in every grid bracket; its minimum is 0.0414 at $s \simeq 0.874587$. The crossing between the trivial floor and the orthogonal-complement floor was bracketed on 401 points and refined by a scalar root solve. These are residual-tested finite diagnostics, not interval-certified bounds, and are not used in the asymptotic theorem.

D Parent cyclic space and endpoint concentration

D.1 Cyclic accessibility

We prove Proposition 8. Because $\mathfrak{A}_{\text{par}}$ is a unital star-algebra, $\mathfrak{A}_{\text{par}}|J_{n,k}\rangle$ is invariant under every element of the algebra and its adjoint, hence is reducing. Every H_a commutes with D_n, and $|J_{n,k}\rangle$ is D_n-fixed, proving the first item.

All rates in Eq. (65) are positive on the connected Johnson graph. The zero eigenvalue of H_a is consequently simple, with ground vector Eq. (69). In finite dimension its spectral projector $P_a = |\psi_a\rangle\langle\psi_a|$

is a polynomial in H_a and belongs to $\mathfrak{A}_{\mathrm{par}}$. Positivity gives $\langle\psi_a|J_{n,k}\rangle > 0$, so $P_a|J_{n,k}\rangle$ is a nonzero multiple of $|\psi_a\rangle$. Finally, both restrictions have this unique ground state, while $\mathcal{K}_J^{\mathrm{par}} \subseteq \mathcal{H}^{D_n}$. Rayleigh–Ritz minimization over the smaller orthogonal complement proves Eq. (74).

D.2 Endpoint concentration

At energy u, occupied and vacant vertices decompose into the same number $k-u$ of positive cyclic runs. Choosing the marked start and the two run compositions gives

$$N_u = \frac{n}{k-u}\binom{k-1}{u}^2, \qquad 0 \le u < k, \tag{94}$$

which is Eq. (75). At $a_f = 7$, the $u = 1$ contribution is $N_1 n^{-7} = n(k-1)n^{-7} = O(n^{-5})$. Consecutive terms $N_u n^{-7u}$ have ratio at most $2k^2 n^{-7} = O(n^{-5})$. Summing the resulting geometric tail proves Eq. (76).

D.3 Adiabatic derivative bounds

A Johnson move removes one selected cycle vertex and inserts one unselected vertex, so $|U(T) - U(S)| \le 2$, and each subset has $k(n-k) = k^2$ neighbours. Differentiating the matrix elements in Eqs. (65) and (68), then bounding the operator norm by the maximum absolute row sum, gives, for constants $c_1, c_2 > 0$ independent of n,

$$\|\partial_s H_{7s}\| \le c_1 n \log n, \qquad \|\partial_s^2 H_{7s}\| \le c_2 n (\log n)^2. \tag{95}$$

Proposition 8 and Eq. (79) give a simple ground state and minimum cyclic gap at least $1024 n^{-13}$.

Apply the quantitative gapped adiabatic bound [14] to $H_{7s}|_{\mathcal{K}_J^{\mathrm{par}}}$; restriction cannot increase either derivative norm. The three contributions scale as $n^{27}\log n$, $n^{27}(\log n)^2$, and $n^{41}(\log n)^2$, divided by T. The last dominates and proves Eq. (80). The final measurement statement follows from Eq. (76) and continuity of measurement probabilities in state distance.

E Normalized comparison proof for the parent-path gap

This appendix fixes the generator convention used in Propositions 9 and 10 and records each comparison factor. For a reversible generator $\mathcal{L}$ with stationary law ν, write

$$\mathcal{E}_{\mathcal{L}}(f,f) = -\langle f, \mathcal{L} f\rangle_\nu, \tag{96}$$

$$\operatorname{gap}(-\mathcal{L}) = \inf_{\operatorname{Var}_\nu(f)>0} \frac{\mathcal{E}_{\mathcal{L}}(f,f)}{\operatorname{Var}_\nu(f)}. \tag{97}$$

Lemma 12 (Three-box heat-bath contraction)**.** *Let $w_0 = \tau$, $w_j = 1$ for $j \ge 1$, with $0 < \tau \le 1$. On weak compositions of any total m into three boxes, let one heat-bath step keep one uniformly chosen coordinate and resample the other two from their conditional product law. The continuous-time generator $\mathcal{L}^*_{3,m}$ obtained by subtracting the identity has gap at least $1/3$.*

E.1 Marked configurations and the cycle zero-range chain

Put $\tau = n^{-a}$ and let $\Omega_{n,k} = \{S \subset \mathbb{Z}_n : |S| = k\}$. Let $\mathcal{Q}_a$ denote the Markov generator with the rates in Eq. (65) and stationary law $\pi_a(S) \propto \tau^{U(S)}$; its symmetric discriminant is H_a. If D_{π_a} is diagonal with entries $\pi_a(S)$, then

$$H_a = -D_{\pi_a}^{1/2} \mathcal{Q}_a D_{\pi_a}^{-1/2}. \tag{98}$$

The unitary map $f \mapsto \sum_S \sqrt{\pi_a(S)} f(S)|S\rangle$ intertwines the D_n actions. Thus the gap of H_a in the fixed space is the Poincaré constant of $\mathcal{Q}_a$ on D_n-invariant observables.

Introduce the marked space

$$\widehat{\Omega}_{n,k} = \{(S,x) : S \in \Omega_{n,k},\ x \in S\}, \tag{99}$$

$$\widehat{\pi}_a(S,x) = \frac{\pi_a(S)}{k}. \tag{100}$$

Lift every Johnson transition that replaces $y \in S$ by $z \notin S$ to the marked space by leaving the mark unchanged when $x \ne y$ and replacing the mark by z when $x = y$. Thus the mark follows its labelled particle. This lifted generator is reversible with respect to $\widehat{\pi}_a$. For the mark-independent lift $\widehat{f}(S,x) = f(S)$, its variance and Dirichlet form are exactly those of f under the original chain.

Starting at the marked particle x, list the occupied vertices in cyclic order and let g_i be the number of vacant vertices after particle i. This gives a bijection

$$(S,x) \longleftrightarrow (x,g), \qquad x \in \mathbb{Z}_n, \tag{101}$$

between marked configurations and anchored weak compositions satisfying

$$g_i \ge 0, \qquad \sum_{i=1}^k g_i = k, \qquad U(g) = \#\{i : g_i = 0\}. \tag{102}$$

Changing the marked particle cyclically shifts g, while rotating or reflecting the subset changes the anchor and possibly the orientation. Here is the first indispensable use of dihedral invariance: only for a D_n-invariant observable $f(S)$ is its marked lift independent of the anchor and representable by a cyclic-and-reflection-invariant function $F(g)$. When the marked particle itself moves, the anchor x changes by the rule above, but F is unaffected by that change. Without D_n invariance the lift would be a function of (x,g) and the composition-only comparison below would not follow.

Define

$$r_a(0)=0, \qquad r_a(1)=\tau, \qquad r_a(j)=1 \quad (j\geq 2). \tag{103}$$

The product weight of a composition is

$$\prod_{i=1}^{k}\prod_{j=1}^{g_i} r_a(j)^{-1} = \tau^{-\#\{i:g_i>0\}} = \tau^{-k}\tau^{U(g)}. \tag{104}$$

Let ν_a be the probability law on weak compositions obtained by normalizing this product weight. The irrelevant factor τ^{-k} shows that the pushforward of the marked law after dropping the anchor is exactly the stationary composition law: every unanchored composition g has precisely n anchored preimages, all with the same weight. In particular,

$$\operatorname{Var}_{\pi_a}(f) = \operatorname{Var}_{\widehat{\pi_a}}(f) = \operatorname{Var}_{\nu_a}(F). \tag{105}$$

Let $P_{C_k}(i,j)=1/2$ when $j=i\pm 1 \pmod k$ and zero otherwise, and let e_i be the ith coordinate vector. We use the Hermon–Salez convention

$$(\mathcal{L}_{P,r}F)(g) = \sum_{i,j} r(g_i)P(i,j)[F(g-e_i+e_j)-F(g)]. \tag{106}$$

The notation $\mathrm{ZRP}(P,r,k)$ below denotes the zero-range generator $\mathcal{L}_{P,r}$ restricted to configurations with k particles. Multiplication by $2/n$ makes every oriented cycle transfer occur at rate $r_a(g_i)/n$. Such a transfer is an adjacent Johnson move. If $g_i=1$ and the receiving box is positive, it creates one new zero and has the Metropolis rate τ/n; if the receiving box is zero, its Metropolis rate is $1/n\geq\tau/n$. For $g_i\geq 2$, the move does not increase U and both rates equal $1/n$. Hence the full Johnson Dirichlet form dominates the local form and, under the lift above,

$$\mathcal{E}_{\mathcal{Q}_a}(f,f) \geq \frac{2}{n}\mathcal{E}_{\mathrm{ZRP}(P_{C_k},r_a,k)}(F,F). \tag{107}$$

The equality of variances is exact; the inequality comes only from dropping nonlocal Johnson edges and, in one local case, lowering a Metropolis rate.

Let $K_k(i,j)=1/k$. Corollary 3 of Hermon and Salez [23], in their site-independent-rate convention, states the pointwise Dirichlet-form comparison

$$\begin{aligned}\mathcal{E}&_{\mathrm{ZRP}(P_{C_k},r_a,k)}(F,F)\\ &\geq [1-\cos(2\pi/k)]\mathcal{E}_{\mathrm{ZRP}(K_k,r_a,k)}(F,F),\end{aligned} \tag{108}$$

because $1-\cos(2\pi/k)$ is the Poincaré constant of P_{C_k} and that of K_k is one. Their comparison applies to every observable before any symmetry restriction and does not require monotone rates. In the present case the rates are nevertheless nondecreasing, $0=r_a(0)\leq r_a(1)=\tau\leq r_a(j)=1$ for $j\geq 2$. The dihedral restriction was already needed to define $F(g)$; restricting the subsequent variational infimum can only increase it. Equations (105)–(108) prove Eq. (77) with its factor $2/n$.

E.2 Three-box contraction

Set $w_0=\tau$ and $w_j=1$ for $j\geq 1$. For $0<z<1$, the probability sequence $p_z(j)\propto w_j z^j$ is log-concave: the only nontrivial inequality is $w_1^2\geq w_0w_2$, which is precisely $\tau\leq 1$. Conditioning two independent variables with this law to have sum s removes the factor z^s and gives

$$\nu_s(u) = \frac{w_u w_{s-u}}{\sum_{v=0}^{s} w_v w_{s-v}}, \qquad 0\leq u\leq s. \tag{109}$$

The discrete Efron monotonicity theorem [24] implies that $U_s\sim\nu_s$ is stochastically nondecreasing in s. Write $\preceq_{\mathrm{st}}$ for stochastic domination. Symmetry under $u\mapsto s-u$ gives, for $t=s+d$,

$$U_s \preceq_{\mathrm{st}} U_t \preceq_{\mathrm{st}} U_s+d. \tag{110}$$

The common-quantile coupling therefore obeys $0\leq U_t-U_s\leq d$ almost surely. The two increments in the coupled splits are U_t-U_s and $d-(U_t-U_s)$, so both are nonnegative.

On the three-box state space use $d_1(x,y)=\frac{1}{2}\sum_i|x_i-y_i|$. Couple two heat-bath steps by holding the same uniformly selected coordinate i. The two resampled pair totals differ by $d=|x_i-y_i|$, and the coupling above gives pairwise ℓ^1 distance d. Together with the held-coordinate distance, the post-update d_1 distance is at most $|x_i-y_i|$. Let W_{1,d_1} denote the 1-Wasserstein distance induced by d_1. Thus the discrete heat-bath kernel P satisfies

$$W_{1,d_1}(\mathsf{P}(x,\cdot),\mathsf{P}(y,\cdot)) \leq \frac{1}{3}\sum_i|x_i-y_i| = \frac{2}{3}d_1(x,y). \tag{111}$$

For a function f, let $\mathrm{Lip}(f)=\sup_{x\neq y}|f(x)-f(y)|/d_1(x,y)$ be its Lipschitz seminorm. It follows that $\mathrm{Lip}(\mathsf{P}f)\leq(2/3)\,\mathrm{Lip}(f)$. Every nonconstant eigenfunction on this finite metric space has positive Lipschitz seminorm, so all nonconstant eigenvalues of the reversible kernel have absolute value at most $2/3$. Therefore $\mathcal{L}^*_{3,m}=\mathsf{P}-I$ has gap at least $1/3$, uniformly in the conserved total m. This proves Lemma 12.

E.3 Caputo recursion with the present normalization

For N boxes and conserved total m, let

$$\Omega_{N,m} = \{g\in\mathbb{Z}^N_{\geq 0} : \textstyle\sum_i g_i = m\}, \tag{112}$$

$$\nu_{N,m}(g) = Z_{N,m}^{-1}\prod_{i=1}^{N} w_{g_i}. \tag{113}$$

Here $Z_{N,m}$ is the normalizing partition function. Let E_{ij} denote conditional expectation under $\nu_{N,m}$ given all coordinates outside the unordered pair $\{i,j\}$. The complete-graph pair heat-bath generator is normalized as

$$\mathcal{L}^{N,m}_{\mathrm{HB}} = \frac{1}{N}\sum_{1\leq i<j\leq N}(E_{ij}-I). \tag{114}$$

For $N = 3$, the discrete kernel $I + \mathcal{L}^{3,m}_{\mathrm{HB}}$ keeps one uniformly chosen coordinate and resamples the remaining pair. It is exactly the kernel of Lemma 12, not a rescaled version.

Choose any $0 < z < 1$ and let the one-box law be $\mu_z(j) \propto w_j z^j$. Then $\nu_{N,m}$ is exactly the product law $\mu_z^{\otimes N}$ conditioned on $\sum_i g_i = m$, because the factor z^m cancels. Consequently, after conditioning on all coordinates outside a set A, the coordinates in A again have their product law conditioned only on the remaining total $m - \sum_{i \notin A} g_i$. This is precisely the "non-interference property" defined before Eq. (2.2) of Caputo [25].

For $m = 0$ the conditioned law is a point mass, contributes no conditional variance, and its gap is assigned the convention $+\infty$. With

$$\underline{\lambda}_3 = \inf_{m \ge 1} \operatorname{gap}(-\mathcal{L}^{3,m}_{\mathrm{HB}}), \tag{115}$$

Theorem 3.1 of Caputo [25], in the normalization (114), gives

$$\operatorname{gap}(-\mathcal{L}^{N,m}_{\mathrm{HB}}) \ge (3\underline{\lambda}_3 - 1)\left(1 - \frac{2}{N}\right) + \frac{1}{N}. \tag{116}$$

Lemma 12 gives $\underline{\lambda}_3 \ge 1/3$. The first term on the right of Eq. (116) is therefore nonnegative, and

$$\operatorname{gap}(-\mathcal{L}^{N,m}_{\mathrm{HB}}) \ge \frac{1}{N} \tag{117}$$

for every conserved total. In the application below, $N = m = k$.

E.4 Conditional two-box comparison

For the mean-field kernel $K_k(i,j) = 1/k$, decompose its zero-range generator into unordered pairs:

$$\mathcal{L}_{\mathrm{ZRP}(K_k, r_a, m)} = \frac{2}{k} \sum_{1 \le i < j \le k} \mathcal{L}^{(2)}_{ij}. \tag{118}$$

Here $\mathcal{L}^{(2)}_{ij}$ is the conditional two-box zero-range generator with kernel K_2, whose two rows are $(1/2, 1/2)$. Thus a unit leaves either member of the pair toward the other at rate $r_a/2$. The prefactor $2/k$ in Eq. (118) converts this to the mean-field rate r_a/k.

Condition on pair total s. For $s \ge 2$, the stationary weights on $\{0, \ldots, s\}$ are τ at the two endpoints and one in the interior, with

$$Z_s = s - 1 + 2\tau. \tag{119}$$

Every nearest-neighbour edge conductance of $\mathcal{L}^{(2)}_{ij}$ is at least $\tau/(2Z_s)$. For the canonical path between $u < v$, Cauchy–Schwarz gives a factor $v - u \le s$. For every edge, the stationary masses on its two sides have product at most $1/4$. Let $\mathcal{E}_s$ be the Dirichlet form of this conditional two-box generator. Consequently

$$\operatorname{Var}_{\nu_s}(f) \le \frac{s}{4} \sum_{u=0}^{s-1} [f(u+1) - f(u)]^2 \le \frac{s Z_s}{2\tau} \mathcal{E}_s(f, f). \tag{120}$$

The conditional pair gap is at least

$$\frac{2\tau}{s Z_s} \ge \frac{2\tau}{s(s+1)} \ge \frac{\tau}{k^2}, \qquad 1 \le s \le k; \tag{121}$$

for $s = 1$ it equals τ, and $s = 0$ has no conditional variance.

Let $\mathcal{E}_{\mathrm{HB}}$ be the form of Eq. (114). Applying Eq. (121) conditionally in every pair and using Eq. (118) gives, with $\mathcal{V}_{ij}(f)$ denoting the expected conditional variance outside $\{i, j\}$,

$$\mathcal{E}_{\mathrm{ZRP}(K_k, r_a, k)}(f, f) \ge \frac{2}{k} \frac{\tau}{k^2} \sum_{i<j} \mathcal{V}_{ij}(f) \tag{122}$$

$$= \frac{2\tau}{k^2} \mathcal{E}_{\mathrm{HB}}(f, f). \tag{123}$$

Combining this with Eq. (117) yields

$$\operatorname{gap} \mathrm{ZRP}(K_k, r_a, k) \ge \frac{2\tau}{k^2} \frac{1}{k} = 2\tau k^{-3}. \tag{124}$$

This displays separately the pair-selection factor $2/k$, the conditional pair estimate τ/k^2, and the heat-bath gap $1/k$.

Finally, for $k \ge 3$, $1 - \cos(2\pi/k) \ge 8/k^2$. Substitution of Eq. (124) into Eq. (77) gives

$$\Delta_{D_n}(a) \ge \frac{2}{n} \frac{8}{k^2} \frac{2\tau}{k^3} = \frac{32\tau}{n k^5} = 1024\, n^{-(a+6)}, \tag{125}$$

because $k = n/2$ and $\tau = n^{-a}$. This completes the proof of Proposition 10. Writing $\Delta_{\mathrm{ZRP}}(a) = \operatorname{gap} \mathrm{ZRP}(K_k, r_a, k)$ and also using cyclic accessibility, the complete protocol-level comparison is

$$\begin{aligned} \Delta^{\mathrm{cyc}}_{\mathrm{par}}(a) &\ge \Delta_{D_n}(a) \\ &\ge \frac{2}{n}\left[1 - \cos\left(\frac{2\pi}{k}\right)\right] \Delta_{\mathrm{ZRP}}(a) \\ &\ge 1024\, n^{-(a+6)}. \end{aligned} \tag{126}$$

F Numerical protocol and full tables

Inputs are frozen with identifiers and SHA-256 hashes. Every reported gap is computed either in the full Hamiltonian or in an explicitly labelled exact orbit or cyclic restriction. Values from different operators are never combined in one unlabeled column. Numerical degeneracy is accepted only relative to eigenpair residuals and matrix scale. Inside a degenerate manifold, let P_0 denote its spectral projector and let $\Pi_{\alpha\beta}$ project onto the joint irrep labelled by (α, β). The irrep content is obtained from

$$\operatorname{rank}(P_0 \Pi_{\alpha\beta} P_0), \tag{127}$$

not by assigning labels to solver-dependent individual eigenvectors.

In the tables below, g_0^{full} and g_0^{acc} are the ranks of the ground projectors on $\mathcal{H}_{\mathrm{addr}}$ and $\mathcal{K}_u$, respectively.

A partition in square brackets labels the corresponding S_k irrep.

For sparse full-space calculations, the central class sum of register transpositions separates every S_k irrep for $k \leq 5$. Acting on a lowest vector with register permutations constructs its full representation multiplet; deflation then tests for further cospectral copies before g_0^{full} is assigned. For the grid row, the two-dimensional global ground manifold is the E irrep of D_4, whereas the accessible branch is A_1. At the analytic hopping-cancellation point, the joint-fixed ground rank agrees with the independently enumerated number of G_B-orbits of optimal covers in every row.

The cyclic basis is constructed independently from the exact potential shells by Eq. (30). Its dimension is stable for relative singular-value tolerances 10^{-9}, 10^{-11}, and 10^{-13} in all eleven rows. The smallest retained singular value is 4.4×10^{-4}, the largest discarded one is 1.1×10^{-15}, and the largest transport-invariance residual is 2.3×10^{-14}. The direct residual of Eq. (52) is at most 2.2×10^{-14} across the audit. The largest orbit-invariance residual is below 5×10^{-13}, and every full-space eigenpair and class-sum residual used below is stored in the machine-readable output. The shell count p in Table 1 ranges from 9 to 31 and is strictly below $\dim \mathcal{K}_u$ in every row; Eq. (33) is thus a structural floor, not a dimension estimate, on this sample. Eight rows are subinstances extracted from the OR-Library Set Cover benchmarks [26, 27]; parent indices, extraction metadata, and hashes are stored in the instance registry.

For the parent family, exact D_n-orbit quotients for $n = 6, 8, \ldots, 18$ place the minimum of a 31-point grid at $a = 7$. Endpoint gaps decrease from 0.166670 at $n = 6$ to 0.0105026 at $n = 18$ and look approximately quadratic on these sizes. This is finite-grid evidence, not a scaling inference; Proposition 10 supplies the uniform certificate.

Table 1: Exact structural reduction for the frozen finite instances. All dimensions refer to the physical, unpadded address space; p is the number of potential shells and $d_{\mathrm{miss}} = \dim \mathcal{H}_{\mathrm{sym}} - \dim \mathcal{K}_u$.

instance	k	$\lvert G_B \rvert$	$\dim \mathcal{H}_{\mathrm{addr}}$	$\dim \mathcal{H}_{\mathrm{sym}}$	p	$\dim \mathcal{K}_u$	d_{miss}	cover orbits
grid-2x4	3	8	512	22	9	22	0	2
rnd-b8-t7-p50-s002	3	1	512	120	9	101	19	12
rnd-b8-t8-p42-s014	4	1	4096	330	15	324	6	7
scpe1-b8-t20	4	1	4096	330	19	328	2	8
scpe1-b8-t30	5	1	32768	792	30	792	0	1
scpe2-b8-t20	4	1	4096	330	19	328	2	3
scpe2-b8-t30	5	1	32768	792	29	792	0	8
scpe3-b8-t20	3	1	512	120	14	119	1	2
scpe3-b8-t30	5	1	32768	792	31	792	0	4
scpe4-b8-t29	5	1	32768	792	30	792	0	1
scpe5-b8-t30	5	1	32768	792	29	792	0	2

Table 2: Problem-endpoint spectra reconstructed from the full Hamiltonian and the exact joint-fixed and cyclic restrictions. Here Δ is the gap above the entire ground manifold, $\delta_{\mathrm{sym}} = E_0^{\mathrm{sym}} - E_0^{\mathrm{full}}$, $\delta_{\mathrm{cyc}} = E_0^{\mathcal{K}} - E_0^{\mathrm{sym}}$, and $p_{\mathrm{opt}}^{\mathrm{min}}$ is minimized over normalized cyclic ground states. The symmetric and cyclic endpoint ground states are unique in every row; the last column is the joint-fixed ground rank at the hopping-cancellation point.

instance	ground S_k content	g_0^{full}	Δ_{full}	δ_{sym}	δ_{cyc}	Δ_{sym}	Δ_{cyc}	$p_{\mathrm{opt}}^{\mathrm{min}}$	$g_0^{\mathrm{sym}}(s_c)$
grid-2x4	$2[3]$	2	0.000382	0.271	0	0.315	0.315	0.9944	2
rnd-b8-t7-p50-s002	$[3] \oplus 2[2,1] \oplus [1,1,1]$	6	0.0563	0	0.0581	0.0581	0.128	0.9986	12
rnd-b8-t8-p42-s014	$[2,1,1]$	3	0.000327	0.0274	0	0.0297	0.0297	0.9955	7
scpe1-b8-t20	$[1,1,1,1]$	1	0.000539	0.0103	0	0.00178	0.00178	0.9983	8
scpe1-b8-t30	$[5]$	1	0.00591	0	0	4.87	4.87	0.9949	1
scpe2-b8-t20	$[2,1,1]$	3	0.00137	0.00674	0	0.18	0.18	0.9966	3
scpe2-b8-t30	$[1,1,1,1,1]$	1	0.00118	0.0316	0	0.0829	0.0829	0.9982	8
scpe3-b8-t20	$[2,1]$	2	0.000658	0.00499	0	0.256	0.256	0.9969	2
scpe3-b8-t30	$[3,1,1]$	6	0.000434	0.00864	0	0.247	0.247	0.9985	4
scpe4-b8-t29	$[5]$	1	0.00608	0	0	4.8	4.8	0.9945	1
scpe5-b8-t30	$[4,1]$	4	0.00306	0.00749	0	0.242	0.242	0.9962	2